

Module 2: Limits, Continuity & Derivatives

Ph.D. Math Camp

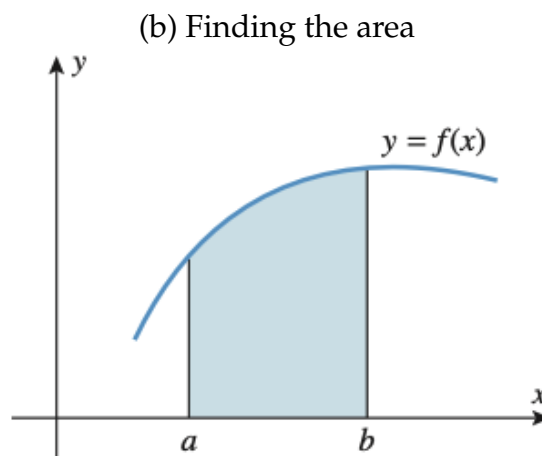
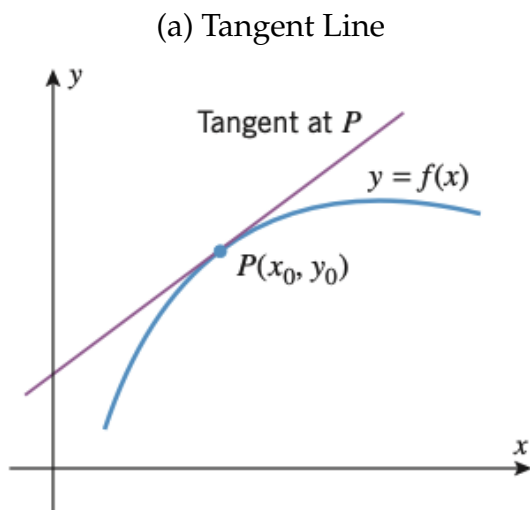
1 Introduction to Calculus

1.1 Limits

What gives calculus its power and distinguishes it from algebra is the concept of the **limit** process.

Many of the ideas of calculus originated with the following two geometric problems:

1. **The Tangent Line Problem:** Given a function f and a point $P(x_0, y_0)$ on its graph, find an equation of the line tangent (just touching) to the graph at $P \rightarrow$ **differential calculus**
2. **The Area Problem:** Given a function f , find the area between the graph of f and an interval $[a, b]$ on the x -axis (especially for curved spaces) $\rightarrow$ **integral calculus**



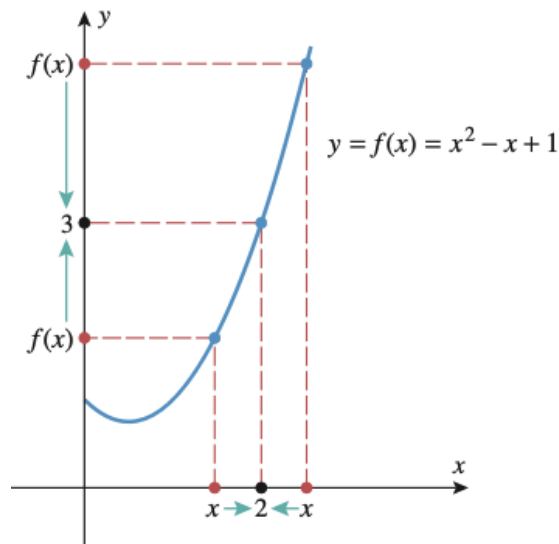
Limits help us answer these questions and apply them in various settings.

Roughly speaking, finding the limit involves determining the value the function $f(x)$ is heading toward as x heads toward c . In this definition, x need not arrive at c , nor must $f(x)$ actually arrive at a destination as x heads toward c . If $f(x)$ is not heading anywhere, the limit does not exist. It's the direction that counts, not the destination.

The most basic use of limits is to describe how a function behaves as the independent variable approaches a given value. For example, let us examine the behavior of the following function for x -values closer and closer to 2:

$$f(x) = x^2 - x + 1$$

It is evident from the graph that the values of $f(x)$ get closer and closer to 3 as values of x are selected closer and closer to 2 on either the left or the right side of 2.



We describe this by saying that the “limit of $f(x) = x^2 - x + 1$ is 3 as x approaches 2 from either side,” and we write

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} (x^2 - x + 1) = 3$$

A (in)formal definition of a limit is as follows:

If the values of $f(x)$ can be made as close as we like to L by taking values of x sufficiently close to a (but not equal to a), then we write

$$\lim_{x \rightarrow a} f(x) = L \quad (1)$$

which is read “the limit of $f(x)$ as x approaches a is L ” or “ $f(x)$ approaches L as x approaches a .” The expression in (1) can also be written as

$$f(x) \rightarrow L \text{ as } x \rightarrow a$$

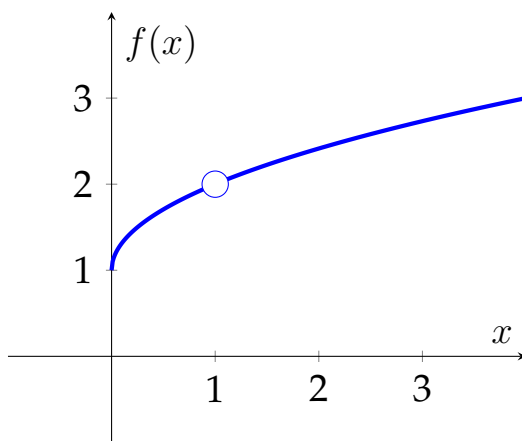
Example: Use numerical evidence to make a conjecture about the value of

$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1}$$

Although the function is undefined at 1, this does not affect the limit.

Table 1: Input and Output Values as x Approaches 1

x	0.99	0.999	0.9999	0.99999	1.00001	1.0001	1.001	1.01
$f(x)$	1.994987	1.999500	1.999950	1.999995	2.000005	2.000050	2.000500	2.004988



The table and the figure show that $f(x)$ gets closer and closer to 2, and hence we conjecture that

$$\lim_{x \rightarrow 1} \frac{x - 1}{\sqrt{x} - 1} = 2$$

Algebraic Properties of Limits

if $\lim_{x \rightarrow c} f(x)$ and $\lim_{x \rightarrow c} g(x)$ exists, then

- $\lim_{x \rightarrow c} k = k$. The limit of a constant, k , is the constant itself.
- $\lim_{x \rightarrow c} [f(x) + g(x)] = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x)$
- $\lim_{x \rightarrow c} [f(x) - g(x)] = \lim_{x \rightarrow c} f(x) - \lim_{x \rightarrow c} g(x)$
- $\lim_{x \rightarrow c} [f(x)g(x)] = [\lim_{x \rightarrow c} f(x)][\lim_{x \rightarrow c} g(x)]$
- $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow c} f(x)}{\lim_{x \rightarrow c} g(x)}$, provided that $\lim_{x \rightarrow c} g(x) \neq 0$
- $\lim_{x \rightarrow c} [f(x)]^p = [\lim_{x \rightarrow c} f(x)]^p$

Examples: Compute the limits for the following

a) $\lim_{x \rightarrow 5} 9 = 9$

b) $\lim_{x \rightarrow 6} x^2 = 36$

c) $\lim_{x \rightarrow 3} 2x^3 = 54$

d) $\lim_{x \rightarrow 2} (x^4 + 3x) = 22$

e) $\lim_{x \rightarrow 4} [(x + 8)(x - 5)] = [\lim_{x \rightarrow 4} (x + 8)][\lim_{x \rightarrow 4} (x - 5)] = -12$

1.2 Limits of Polynomials and Rational Functions

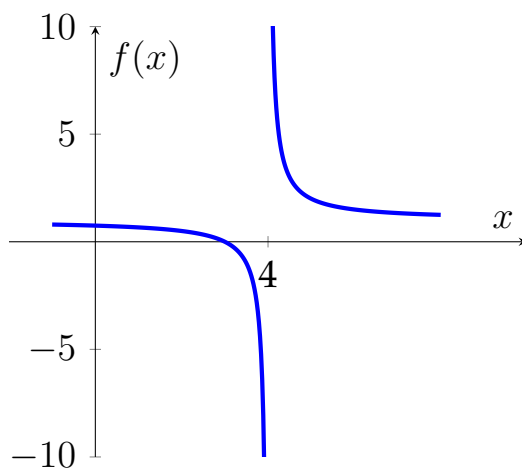
If $p(x)$ and $q(x)$ are polynomials, then

$$\lim_{x \rightarrow c} p(x) = p(c) \quad \text{and}$$

$$\lim_{x \rightarrow c} \frac{p(x)}{q(x)} = \frac{p(c)}{q(c)} \quad \text{and} \quad q(c) \neq 0$$

A key point to remember is that *if the denominator of a given rational function approaches zero and the numerator does not, the limit does not exist.*

Take $\lim_{x \rightarrow 4} \frac{x-3}{x-4}$. Here if we plug in 4 we get $\frac{1}{0}$. We can maybe see why in the figure.



This function goes both to positive and negative ∞ as you approach 4. So, it is not clear what the limit is. We will return to this briefly.

When both the numerator and the denominator approach zero, you need to simplify the function algebraically to find if the limit exists and if so, its value.

Examples: Please find

$$\text{a) } \lim_{x \rightarrow 7} \frac{x-7}{x^2-49} = \frac{0}{0}$$

$$\lim_{x \rightarrow 7} \frac{x-7}{x^2-49} = \lim_{x \rightarrow 7} \frac{x-7}{(x-7)(x+7)} = \lim_{x \rightarrow 7} \frac{\cancel{(x-7)}}{\cancel{(x-7)}(x+7)} = \lim_{x \rightarrow 7} \frac{1}{(x+7)} = \frac{1}{7+7} = \frac{1}{14}$$

$$\text{b) } \lim_{x \rightarrow 6} \frac{x^2-2x-24}{x-6}$$

$$\lim_{x \rightarrow 6} \frac{x^2-2x-24}{x-6} = \lim_{x \rightarrow 6} \frac{\cancel{(x-6)}(x+4)}{\cancel{(x-6)}} = \lim_{x \rightarrow 6} (x+4) = 10$$

One-Sided Limits

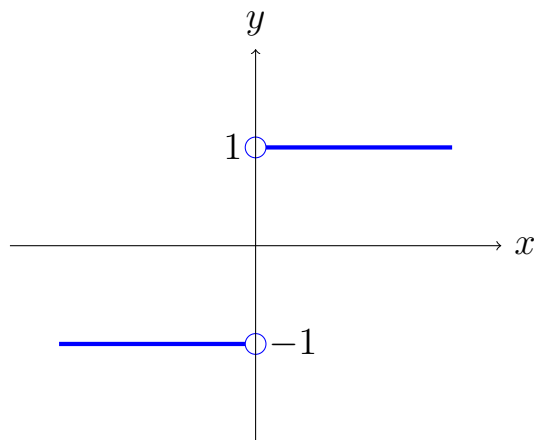
Recall issue we had with $\lim_{x \rightarrow 4} \frac{x-3}{x-4}$. Thus far, we have discussed **two-sided limits** because it requires the values of $f(x)$ to get closer and closer to L as values of x are taken from either side of $x = a$. However, some functions exhibit different behaviors on the two sides of an x -value a , in which case it is necessary to distinguish whether values of x near a are on the left side or on the right side of a for purposes

of investigating limiting behavior.

Informal definition: If $f(x)$ approaches L as x tends towards c from the left ($x < c$), we write $\lim_{x \rightarrow c^-} f(x) = L$. Likewise, if $f(x)$ approaches M as x tends toward c from the right side ($x > c$), then $\lim_{x \rightarrow c^+} f(x) = M$. With this notation, the superscript “+” indicates a limit from the right and the superscript “-” indicates a limit from the left.

Consider the function

$$f(x) = \frac{|x|}{x} = \begin{cases} 1, & \text{if } x > 0 \\ -1, & \text{if } x < 0 \end{cases}$$



Evaluate the one-sided limits $\lim_{x \rightarrow 0^-} f(x)$ and $\lim_{x \rightarrow 0^+} f(x)$. Notice that the two-sided limit $\lim_{x \rightarrow 0} f(x)$ does not exist because the functional values $f(x)$ do not approach a single value L .

In general, the criterion for the existence of a limit can be stated as:

The two-sided limit $\lim_{x \rightarrow a} f(x) = L$ exists if $\lim_{x \rightarrow a^+} f(x) = \lim_{x \rightarrow a^-} f(x) = L$

1.3 Limits Involving Infinity

If $f(x)$ approaches the number L as x increases without bound or the number M as x decreases without bound, we write

$$\lim_{x \rightarrow +\infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow -\infty} f(x) = M$$

The behavior of a function $f(x)$ as x increases without bound or decreases without bound is sometimes called the end behavior of the function.

Sometimes, taking a limit results in unbounded growth or decline. Here we get situations like the following

$$\lim_{x \rightarrow c^-} f(x) = +\infty \quad \text{or} \quad \lim_{x \rightarrow c^+} f(x) = -\infty$$

Limits at Infinity involving powers and reciprocal powers can be computed as follows:

For $n > 0$

- $\lim_{x \rightarrow +\infty} x^n = +\infty$
- $\lim_{x \rightarrow +\infty} -x^n = -\infty$

For $\{n | n \text{ is an odd integer}\}$

- $\lim_{x \rightarrow -\infty} x^n = -\infty$

For $\{n | n \text{ is an even integer}\}$

- $\lim_{x \rightarrow -\infty} x^n = +\infty$
- $\lim_{x \rightarrow \pm\infty} \frac{1}{x^n} = 0$

The limit at infinity of a polynomial is determined by the highest power term of the polynomial.

Specifically, if $a_n \neq 0$, then: $\lim_{x \rightarrow +\infty} (a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0) = \lim_{x \rightarrow +\infty} a_n x^n$

That is, to find the limit at infinity of a polynomial, take the limit of the term of the highest degree.

Example:

a) $\lim_{x \rightarrow +\infty} (3 - x^2 + 6x^4) = +\infty$

b) $\lim_{x \rightarrow +\infty} (3 - x^2 - 6x^4) = -\infty$

Limits at Infinity of Rational Functions

To find the limit at infinity of a rational function

1. Compare the degrees of the numerator and the denominator and divide the numerator and the denominator by x raised to the smaller of those degrees.
2. Take the limits of the new numerator and denominator

Note: to find a limit at infinity of a rational function we CANNOT just evaluate the highest term.

Some shortcuts for finding limits at infinity of rational functions:

- If the degree of the numerator is equal to the degree of the denominator, the **limit equals the ratio of the leading coefficients.**
- If the degree of the numerator is less than the degree of the denominator, the limit is zero.
- If the degree of the numerator is greater than the degree of the denominator, the limit is infinity or negative infinity depending on the sign and power of the numerator

Examples: Please find

$$\begin{aligned} \text{a) } \lim_{x \rightarrow +\infty} \frac{3x^2 - 7x}{4x^2 - 21} &= \lim_{x \rightarrow +\infty} \frac{\frac{3x^2}{x^2} - \frac{7x}{x^2}}{\frac{4x^2}{x^2} - \frac{21}{x^2}} = \lim_{x \rightarrow +\infty} \frac{3 - \frac{7}{x}}{4 - \frac{21}{x^2}} = \frac{3}{4} \end{aligned}$$

$$\text{b) } \lim_{x \rightarrow +\infty} \frac{4x^5}{x^3 + 2x} = +\infty$$

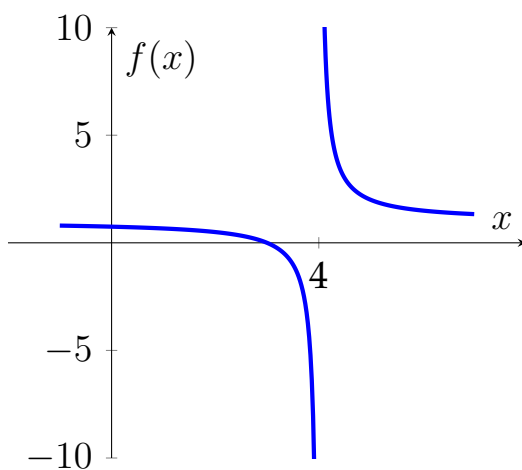
$$\text{c) } \lim_{x \rightarrow +\infty} \frac{-x^3 + 2x + 1}{x - 3} = -\infty$$

2 Introduction to Calculus (continued)

2.1 Asymptotes

Limits involving infinity can be used to describe graphical features called **asymptotes**. In particular, the graph of a function $f(x)$ is said to have a **vertical asymptote** at $x = c$ if $f(x)$ increases or decreases without bound as x tends towards c , from either the right or the left.

Recall, $\lim_{x \rightarrow 4} \frac{x-3}{x-4}$



As x approaches 4 from the left, the functional values decrease without bound, but they increase without bound if the approach is from the right.

The concept of one-sided limit summarizes this behavior as

$$\lim_{x \rightarrow 4^-} \frac{x-3}{x-4} = -\infty \quad \text{and} \quad \lim_{x \rightarrow 4^+} \frac{x-3}{x-4} = +\infty$$

More formally, The line $x = c$ is a *vertical asymptote* of the graph of $f(x)$ if either

$$\lim_{x \rightarrow c^-} f(x) = \pm\infty \quad \text{or} \quad \lim_{x \rightarrow c^+} f(x) = \pm\infty$$

In general, a rational function $R(x) = \frac{p(x)}{q(x)}$ has a vertical asymptote $x = c$ whenever $q(c) = 0$ but $p(c) \neq 0$.

Examples: Determine all vertical asymptotes

a) $f(x) = \frac{x^2-9}{x^2+3x}$ Set denominator to zero and solve $\implies$ vertical asymptote at $x = 0$. At $x = -3$ the numerator is also 0 (removable hole).

b) $g(x) = \frac{3x^3-6x^2}{x^2-4}$. Factoring the denominator (difference of squares) leads to $(x+2)(x-2) = 0 \implies x = -2$ is a vertical asymptote. At $x = 2$ the numerator is also 0 (hole).

When a function $f(x)$ tends towards a finite value b as x either increases or decreases without bound (or both), then the horizontal line $y = b$ is called the **horizontal asymptote** of the graph of $f(x)$.

Definition: The horizontal line $y = b$ is called the **horizontal asymptote** of the graph if

$$\lim_{x \rightarrow -\infty} f(x) = b \text{ or } \lim_{x \rightarrow +\infty} f(x) = b$$

Examples: Determine all vertical and horizontal asymptotes

a) $f(x) = \frac{3x^2}{x^2+2x-15}$.

To find Horizontal asymptote $\lim_{x \rightarrow +\infty} \frac{3x^2}{x^2+2x-15} \implies \lim_{x \rightarrow +\infty} \frac{3x^2}{x^2} = 3$

To find vertical asymptote set the denominator to zero $x^2 + 2x - 15 = 0 \implies (x+5)(x-3) = 0 \implies x = -5; x = 3$

3 Continuity

Informally, a **continuous function** is one whose graph can be drawn without a pencil leaving the paper; that is, a graph with no holes or gaps.

More formally, a function f is **continuous** at a point c if

- $f(c)$ is defined
- $\lim_{x \rightarrow c} f(x)$ exists
- $\lim_{x \rightarrow c} f(x) = f(c)$

If each condition is met, then the function is continuous at c . If $f(x)$ is not continuous at c , it is said to have a **discontinuity** there. Polynomial or rational functions are continuous **whenever they are defined**.

Examples: Functions with discontinuities

a) $f(x) = \frac{1}{x}$ at $x=0$

b) $f(x) = \frac{x-3}{x-4}$ at $x=4$

Examples: Are the following functions continuous at the specified values

1. Is $f(x) = 2x^2 + x - 2$ continuous at $x = 4$?

2. Is $f(x) = \frac{x-4}{x+5}$ continuous at $x = 2$?

3. Is $g(x) = \frac{x^2-4}{x-2}$ continuous at $x = 2$?

4. Is $g(x) = \frac{x^3-8}{x-2}$ continuous at $x = 2$?

Examples:

a) Is $f(x) = 2x^2 + x - 2$ continuous at $x = 4$?

(i) $f(x)$ defined? $f(4) = 2(4)^2 + 4 - 2 = 34$ ✓

(ii) $\lim_{x \rightarrow 4} f(x) = 34$ ✓

(iii) $\lim_{x \rightarrow 4} f(x) = f(4)$ ✓

b) Is $f(x) = \frac{x-4}{x+5}$ continuous at $x = 2$?

(i) $f(x)$ defined? $f(2) = \frac{-2}{7}$ ✓

(ii) $\lim_{x \rightarrow 2} f(x) = \frac{-2}{7}$ ✓

$$(iii) \lim_{x \rightarrow 2} f(x) = f(2) \quad \checkmark$$

c) Is $g(x) = \frac{x^2-4}{x-2}$ continuous at $x = 2$?

$$(i) f(x) \text{ defined? } f(2) = \frac{0}{0} \quad \times$$

$$(ii) \lim_{x \rightarrow 2} f(x) = 4 \quad \checkmark$$

$$(iii) \lim_{x \rightarrow 2} f(x) \neq f(2) \quad \times$$

d) Is $g(x) = \frac{x^3-8}{x-2}$ continuous at $x = 2$?

$$(i) f(x) \text{ defined? } f(2) = \frac{0}{0} \quad \times$$

$$(ii) \lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{\cancel{(x-2)}(x^2+2x+4)}{\cancel{(x-2)}} = 12 \quad \checkmark$$

$$(iii) \lim_{x \rightarrow 2} f(x) \neq f(2) \quad \times$$

Continuity of a Function on an Interval

For many applications of calculus, it is useful to have the following definitions of continuity on open and closed intervals.

1. A function f is said to be continuous on the open interval $a < x < b$ if it is continuous at each point x in the interval.
2. It is said to be continuous on the closed interval $a \leq x \leq b$ if f is continuous on the open interval $a < x < b$ and $f(x)$ approaches $f(a)$ as x approaches a from the right and $f(x)$ approaches $f(b)$ as x approaches b from the left.

In simpler terms– continuity on an interval means the graph of f is “one piece” throughout the interval.

¹To factor $x^3 - 8$, we can use the difference of cubes formula, which states that $a^3 - b^3$ can be factored as $(a - b)(a^2 + ab + b^2)$. In this case, $a = x$ and $b = 2$:

$$x^3 - 8 = (x - 2)(x^2 + 2x + 4)$$

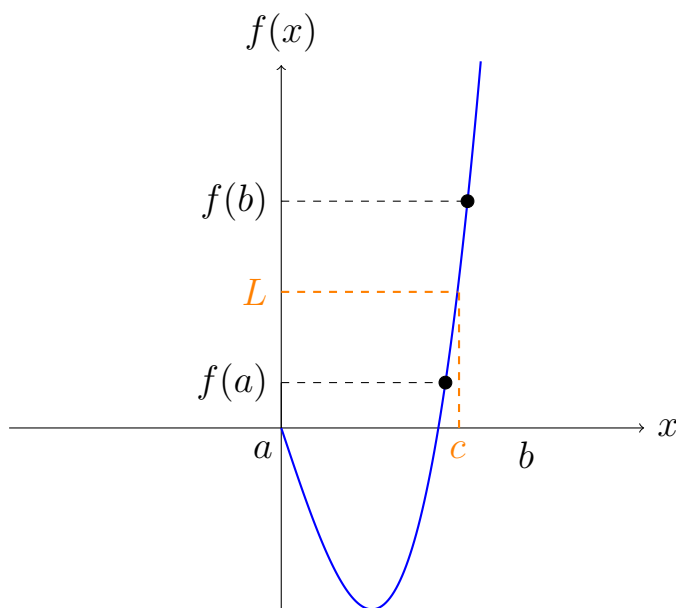
Example: Discuss the continuity of the function $f(x) = \frac{x+2}{x-3}$ on the open interval $-2 < x < 3$ and on the closed interval $-2 \leq x \leq 3$

The function is continuous everywhere except 3. Thus, it is continuous within the open interval. However, $f(3)$ is undefined and the limit as it approaches 3 is infinite. Thus, not continuous on closed interval.

3.0.1 The Intermediate Value Property

An important feature of continuous functions on a closed interval is the **intermediate value property**. In other words, a *continuous function attains all values between any two of its values*.

More formally, the **intermediate value property** says that if f is continuous on the interval $a \leq x \leq b$ and L is a number between $f(a)$ and $f(b)$, then $f(c) = L$ for some number c between a and b .



Example: Show that the equation $x^2 - x - 1 = \frac{1}{x+1}$ has a solution for $1 \leq x \leq 2$.

Rewrite $f(x) = x^2 - x - 1 - \frac{1}{x+1} = 0$. Only place not continuous is $x = -1$.

$$\implies f(1) = 1 - 1 - 1 - \frac{1}{2} = -\frac{3}{2} \quad \text{and} \quad f(2) = 4 - 2 - 1 - \frac{1}{3} = \frac{2}{3}$$

Since f is continuous on $[1, 2]$ (its only discontinuity, $x = -1$, lies outside the interval) and $f(1) < 0 < f(2)$, the intermediate value property guarantees a solution $f(c) = 0$ for some c in the interval $1 \leq x \leq 2$.

4 Derivatives

4.1 Rules

Returning to the tangent problem. Suppose that we wanted an equation to the line that is tangent to $y = f(x)$ at point P in Figure 2. What would we need? Maybe we could use the point-slope formula.

$$y - y_0 = m(x - x_0)$$

We could rewrite this as

$$y - f(x_0) = m(x - x_0)$$

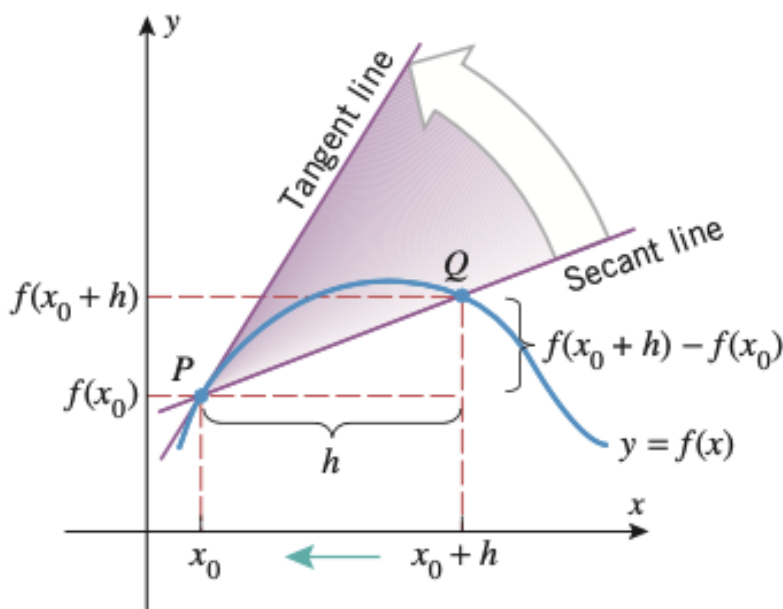
Tangency $\implies$ Just touches the curve at a point and the tangent line and curve have the same slope.

We have our point, P , which is $(x_0, f(x_0))$, but what about m ? We could try to calculate the slope but we only have one point P , and the slope formula requires two points, $\frac{y_1 - y_0}{x_1 - x_0} = \frac{f(x_1) - f(x_0)}{x_1 - x_0} = \frac{y_2 - y_1}{x_2 - x_1}$

Assuming we don't know better, we could try to approximate the slope using a **secant** line that cuts the curve in two places so that we can calculate the slope.²

²A secant line is a straight line that intersects a curve at two points

Figure 2: Tangent + Secant



Here, we can use the two points on the secant line to get a poor approximation of the slope.

Point 1: $x_0, f(x_0) \implies (x_1, y_1)$

Point 2: $x_0 + h, f(x_0 + h) \implies (x_2, y_2)$

$$m_{sec} = \frac{f(x_0 + h) - f(x_0)}{(x_0 + h) - x_0}$$

We can simplify the denominator

$$m_{sec} = \frac{f(x_0 + h) - f(x_0)}{h}$$

Suppose the true slope at point P is m_{tan} . It is clear that $m_{sec} < m_{tan}$ since the secant line is flatter.

Could we do a better job with a second point closer to x_0 (i.e., smaller h)? Yes! We could pick a smaller and smaller h until we get closer and closer to x_0 . In effect, to get m_{tan} we would need to take a limit as $h \rightarrow 0$.

Thus to get the slope m_{tan} at P we can take the limit of the secant equation as $h \rightarrow 0$

$$m_{tan} = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h}$$

Note that this limit may not exist. If it does, it is the slope of the tangent line at $(x_0, f(x_0))$. We call this the first derivative of f with respect to x and can denote it as $f'(x)$.

Formally, the limit definition of a derivative is

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

Now, when we want to find the slope or derivative of a non-linear function the slope will be changing as you move across the curve. Thus, typically, the derivative will result in a formula (another equation) that we can use to calculate the slope at any given point.

Example: Using the limit definition of a derivative find the slope of the functions

1. $f(x) = x^2 + 2x + 7$ at $x = 2$.

2. $f(x) = \sqrt{x}$ at $x = 4$

1. $f(x) = x^2 + 2x + 7$ at $x = 2$.

$$f'(x) = \lim_{h \rightarrow 0} \frac{(x + h)^2 + 2(x + h) + 7 - (x^2 + 2x + 7)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 + 2x + 2h + 7 - x^2 - 2x - 7}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 + 2h}{h}$$

$$= \lim_{h \rightarrow 0} (2x + h + 2)$$

$$= 2x + 2 \Big|_{x=2} = 6$$

2. $f(x) = \sqrt{x}$ at $x = 4$

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\sqrt{x+h} - \sqrt{x}}{h} &= \frac{\sqrt{x+h} - \sqrt{x}}{h} \left(\frac{\sqrt{x+h} + \sqrt{x}}{\sqrt{x+h} + \sqrt{x}} \right) = \frac{x+h-x}{h(\sqrt{x+h} + \sqrt{x})} \\ &= \frac{1}{2\sqrt{x}} \Big|_{x=4} = \frac{1}{2\sqrt{4}} = \frac{1}{4} \end{aligned}$$

We say that $f(x)$ is *differentiable* at c if $f'(c)$ exists i.e., the limit exists as $x \rightarrow c$.

Another way to interpret the slope or f' is that the quantity f changes at the rate $f'(c)$ with respect to x when $x = c$. The derivative tells us the **rate of change** of the function with respect to x . We can also express the derivative of f as $\frac{df}{dx}$.

Techniques of Differentiation

The following are handy rules for finding a derivative and will be very useful in your coursework.

- **The constant rule:** For any constant c , $\frac{d}{dx}(c) = 0$. The derivative of a constant $f(x) = c$ is zero, it is a flat line with no slope.
- **The Power Rule:** For any real number n ,

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Reduce the exponent n of x by 1 and multiply by the original exponent.

- **The constant Multiple Rule:** If c is a constant and f is differentiable, then so is the function cf and

$$\frac{d}{dx}cf(x) = c\frac{df}{dx}$$

- **The sum rule:** If $f(x)$ and $g(x)$ are differentiable, then so is the sum $S(x) = f(x) + g(x)$ and $S'(x) = f'(x) + g'(x)$; that is the derivative of a sum is the sum of the derivatives.

$$\frac{d}{dx}[f(x) + g(x)] = \frac{df}{dx} + \frac{dg}{dx}$$

- **The product rule:** If $f(x)$ and $g(x)$ are differentiable at x , then so is their product $P(x) = f(x)g(x)$. Thus,

$$\frac{d}{dx}[f(x)g(x)] = \frac{df}{dx} \cdot g(x) + \frac{dg}{dx} \cdot f(x)$$

or equivalently.

$$(fg)' = gf' + fg'$$

In words, the derivative of the product is “the derivative of the first times the second plus the derivative of the second times the first.”

- **The quotient rule:** If $f(x)$ and $g(x)$ are differentiable functions, then so is the quotient $Q(x) = \frac{f(x)}{g(x)}$

$$\frac{d}{dx} \left(\frac{f(x)}{g(x)} \right) = \frac{g(x)\frac{df}{dx} - f(x)\frac{dg}{dx}}{g^2(x)}$$

In words, the derivative of the quotient is “the derivative of the top times the bottom minus the derivative of the bottom times the top all over the bottom squared.”

Note that the quotient rule should not be used if you can easily use the power rule and product rule

$$y = \frac{2}{x^3} - \frac{x}{3} - \frac{4}{5} + \frac{x+1}{x}$$

Examples:

a) $y = 8x^3$

b) $f(x) = -6x^5$

c) $f(x) = 5x^2$

d) $y = \frac{7}{x}$

e) $f(x) = 18\sqrt{x}$

f) $f(x) = 5x^4(3x - 7)$

g) $f(x) = 7x^5 + x^3$

h) $f(x) = 5x^4 - (3x - 7)$

i) $f(x) = 7x^9(3x^2 - 12)$

j) $y = \frac{10x^8 - 6x^7}{2x}$

k) $y = \frac{3x^8 - 4x^7}{4x^3}$

- **The chain rule:** Suppose that $y = f(u)$ is a differentiable function of u , and $u = g(x)$ is a differentiable function of x . Then that means $y = f(g(x))$ is a composite function $f \circ g$ at x and

$$\frac{dy}{dx} = \frac{df}{du} \frac{du}{dx}$$

That is, the derivative with respect to x is the derivative of f with respect to its argument u times the derivative of g with respect to its argument x .

Practically, in functional form, we can write the chain rule as

$$\frac{d}{dx} (f(g(x))) = g'(x)f'[g(x)]$$

In words, the derivative of the inside times the derivative of the outside (applied to the inside).

Another version of the chain rule applies to the following case (the general power rule)

$$\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1} \frac{d}{dx}[g(x)]$$

Examples: Find derivatives using chain rule

a) $f(x) = (3x^4 + 5)^6$

b) $y = \frac{1}{7x^3 + 13x^2 + 3}$

c) $y = \sqrt{34 - 6x^2}$

d) $y = \frac{(x+2)^2 \sqrt{x}}{2x}$

Examples: Let's revisit our tangent line and use our derivative rules to find the equation of the line tangent to a curve.

1. Find the equation of the line tangent to the curve $y = x^3$ at the point where $x = -1$
2. Find the equation of the line tangent to the curve $y = (2x + 1)(x^2 - x + 3)$ at point $x = 0$.

- a) Find the equation of the line tangent to the curve $y = x^3$ at the point where $x = -1$

Find $\frac{dy}{dx} = 3x^2$, the slope formula.

Find the point of the tangency $y(-1) = -1 \implies (-1, -1)$. What is the slope at $x = -1$? $\frac{dy}{dx} = 3x^2 \Big|_{x=-1} = 3$

Use point slope formula $(y - (-1)) = 3(x - (-1)) \implies y + 1 = 3x + 3$

$$\implies \boxed{y = 3x + 2}$$

- b) Find the equation of the line tangent to the curve $y = (2x + 1)(x^2 - x + 3)$ at point $x = 0$.

4.2 Implicit Differentiation

So far, the functions we have worked with have all been given by equations of the form $y = f(x)$ in which the dependent variable y on the left is given explicitly by an expression on the right involving the independent variable x . That is, the function is explicitly given on the right-hand side of the equation. A function in this form is said to be in explicit form.

However, sometimes we will need to be able to take the derivative of a function of **implicit form**. In this form, the dependent variable has not been written explicitly in terms of the independent variable.

Example: Label the following as explicit or implicit

a) $y = \frac{x^{11} - x^4 + 3}{x^2} \rightarrow \text{Explicit}$

b) $x^2y^5 = 2xy + y^3x \rightarrow \text{Implicit}$

Suppose you have an equation that defines y implicitly as a function of x , and you want to find the derivative $\frac{dy}{dx}$.

- One possibility is that you may be able to solve the equation for y explicitly in terms of x . In such cases, we can determine the derivative explicitly.
- More often than not, this may be impossible. Thus, we need a technique that allows us to solve for $\frac{dy}{dx}$ without having to solve for y explicitly.

The procedure for doing so is called **implicit differentiation**.

Suppose an equation defines y implicitly as a differentiable function of x . To find $\frac{dy}{dx}$

1. Differentiate both sides of the equation with respect to x , keeping in mind that y is a function of x that is not explicitly stated, and is a function of x . Thus, we must use the chain rule when differentiating terms containing $y = y(x)$.
2. Solve the differentiated equation algebraically for $\frac{dy}{dx}$. This expression is not explicit but involves x and y .

In implicit differentiation, you may find it helps to rewrite the original equation substituting " $y(x)$ " for " y " when differentiating.

Let's go through a problem. Say we want to find $\frac{dy}{dx}$ for $3x^4 - 7y^5 = 86$.

Differentiate with respect to x to get

$$12x^3 \frac{dx}{dx} - 7(5y^4 \frac{dy}{dx}) = \frac{d}{dx} 86^0$$

Simplifying we get,

$$12x^3 - 35y^4 \frac{dy}{dx} = 0 \implies 12x^3 = 35y^4 \frac{dy}{dx} \implies \boxed{\frac{dy}{dx} = \frac{12x^3}{35y^4}}$$

Examples: Please find $\frac{dy}{dx}$

(i) $x^4 y^6 + 3y = 11,664$

(ii) $2x^3 + 5xy + 6y^2 = 13$

(iii) $(2x^3 + 7y)^2 = x^5$

4.3 Differentiation of Logarithmic and Exponential Functions

4.4 Exponential Functions

Let's find the derivative of $f(x) = e^x$ using the limit definition of a derivative.

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

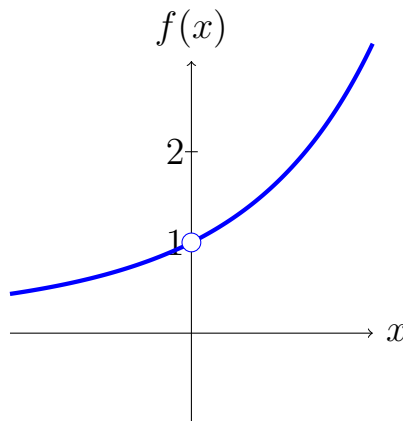
$$\Rightarrow \lim_{h \rightarrow 0} \frac{e^{(x+h)} - e^x}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{e^x e^h - e^x}{h}$$

$$\Rightarrow \lim_{h \rightarrow 0} \frac{e^x(e^h - 1)}{h}$$

$$\Rightarrow e^x \cdot \lim_{h \rightarrow 0} \frac{(e^h - 1)}{h}$$

What is $\lim_{h \rightarrow 0} \frac{(e^h - 1)}{h}$? Let's try to find this graphically by taking an aside and rewriting the function as $\frac{(e^x - 1)}{x}$.



What is the limit as $x \rightarrow 0$? It is 1.

$$\implies \frac{d}{dx} e^x = e^x \cdot \lim_{h \rightarrow 0} \frac{(e^h - 1)}{h} = e^x$$

Thus the derivative of e^x is itself. Geometrically this means that at each point $P(c, e^c)$ on the curve $y = e^x$, the slope is equal to e^c , the y coordinate of P .

The Chain Rule for Exponential Functions

If $u(x)$ is a differentiable function of x , then

$$\frac{d}{dx} [e^{u(x)}] = e^{u(x)} \frac{du}{dx}$$

Examples: Differentiate the following

(i) $f(x) = e^{x^2}$

(ii) $f(x) = 3e^{7-2x}$

(iii) $y = 3xe^{2x}$

Logarithmic Functions

Let's now find the derivative of the $\ln x$ function.

Let $g(x) = \ln(x)$. Thus, if we exponentiate both sides we get $e^{g(x)} = x$

Let's take the derivative of both sides with respect to x (use chain rule on LHS)

$$g'(x)e^{g(x)} = 1. \text{ We know that } e^{g(x)} = x$$

$$g'(x) x = 1$$

$$\implies g'(x) = \frac{1}{x}$$

$$\implies g'(x) = \boxed{\frac{d}{dx} \ln(x) = \frac{1}{x}}, x > 0$$

Thus, $\frac{d}{dx} \ln(x) = \frac{1}{x}$ for $x > 0$.

Examples:

Chain Rule for Logs

If $u(x)$ is a differentiable function of x , then

$$\frac{d}{dx} [\ln u(x)] = \frac{1}{u(x)} \frac{du}{dx} \text{ or, equivalently, } \frac{u'(x)}{u(x)}$$

Examples:

(i) $f(x) = x \ln(x)$

(ii) $f(x) = \frac{\ln(\sqrt[3]{x^2})}{x^4}$ (use Power Rule for Logarithms)

(iii) $y = 3xe^{2x}$

(iv) $f(x) = \ln 6x^2$

(v) $f(x) = \ln(x^2 + 6x + 2)$

(vi) $y = \ln(1 + x)$

(vii) $y = 3 \ln(1 + x)^2$

Sometimes you can simplify differentiation if you first take a log.

Examples:

$$(i) \ g(x) = \frac{(3x^5-4)(2x^3+9)}{(7x^4-5)}$$

$$\ln(g(x)) = \ln((3x^5 - 4)(2x^3 + 9)) - \ln(7x^4 - 5)$$

$$\ln(g(x)) = \ln(3x^5 - 4) + \ln(2x^3 + 9) - \ln(7x^4 - 5)$$

$$\frac{d}{dx} \ln(g(x)) = \frac{15x^4}{3x^5 - 4} + \frac{6x^2}{2x^3 + 9} - \frac{28x^3}{7x^4 - 5}$$

$$\frac{g'(x)}{g(x)} = \frac{15x^4}{3x^5 - 4} + \frac{6x^2}{2x^3 + 9} - \frac{28x^3}{7x^4 - 5}$$

$$\implies g'(x) = \left(\frac{15x^4}{3x^5 - 4} + \frac{6x^2}{2x^3 + 9} - \frac{28x^3}{7x^4 - 5} \right) \cdot g(x)$$

$$g'(x) = \left(\frac{15x^4}{3x^5 - 4} + \frac{6x^2}{2x^3 + 9} - \frac{28x^3}{7x^4 - 5} \right) \left(\frac{(3x^5 - 4)(2x^3 + 9)}{(7x^4 - 5)} \right)$$

$$(ii) \ g(x) = \frac{(5x^3-8)(3x^4+7)}{(9x^5-2)}$$

$$(iii) \ g(x) = (x^3 - 2)(x^2 - 3)(8x - 5)$$

$$(iv) \ y = x^x$$

$$\ln y(x) = \ln x^x \implies \ln y(x) = x \ln x$$

$$\implies \frac{1}{y(x)} \frac{dy}{dx} = \ln x + \overset{1}{\cancel{x}}$$

$$\implies \frac{dy}{dx} = (\ln x + 1) \cdot y(x)$$

$$\implies \frac{dy}{dx} = (\ln x + 1) \cdot x^x$$

5 L'Hôpital's Rule

L'Hôpital's Rule is useful when evaluating limits involving fractions where both the numerator and the denominator approach zero or infinity. The rule allows us to find the limit of such expressions by taking the derivative of the numerator and denominator separately and then reevaluating the limit.

Suppose we have a limit of the form $\lim_{x \rightarrow c} \frac{f(x)}{g(x)}$, where both $f(x)$ and $g(x)$ approach zero or infinity as x approaches c , and both functions are differentiable in some open interval containing c , except possibly at c itself. If the limit of $\frac{f'(x)}{g'(x)}$ exists (or is infinite) as x approaches c , then the original limit is equal to the limit of $\frac{f'(x)}{g'(x)}$ as x approaches c . In other words, if the original limit is in an indeterminate form (e.g., $\frac{0}{0}$ or $\frac{\infty}{\infty}$), we can use L'Hôpital's Rule to evaluate.

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

For example, consider the limit $\lim_{x \rightarrow 0} \frac{e^x - 1}{x}$. If we try to substitute $x = 0$ directly, we get $\frac{0}{0}$, which is an indeterminate form.

In such cases, L'Hôpital's Rule provides a way to find the limit by taking the derivatives of $e^x - 1$ and x separately and then reevaluating the limit. Applying L'Hôpital's Rule to this example, we find that the limit is equal to 1.

$$\lim_{x \rightarrow 0} \frac{e^x - 1}{x} = \lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)} = \lim_{x \rightarrow 0} \frac{e^x}{1} = e^0 = 1$$

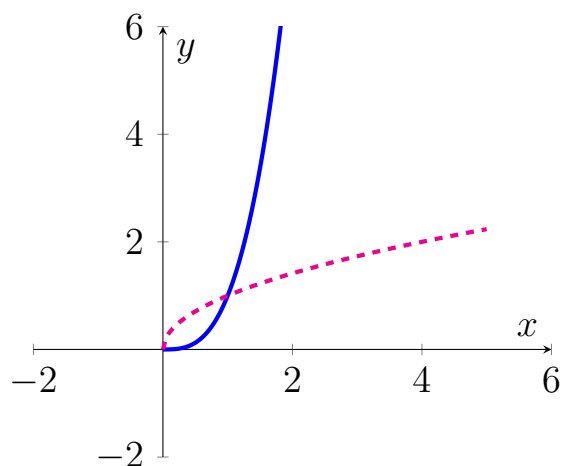
6 Second Derivatives

The **second derivative** tells us the rate of change of the rate of change (or the rate of change in the slope). So, the second derivative of f at x is written as $f''(x)$ or $\frac{df'(x)}{dx}$ or $\frac{d^2 f(x)}{dx^2}$.

You get the second derivative by taking a derivative of the new function created that we called the first derivative.

This concept comes up repeatedly in public policy settings. For instance, demographers are interested in population growth and the change in the rate at which

population growth (or decline) is taking place. Is it increasing at an increasing rate? Or is it increasing, but leveling off?



$f''(x)$ is the derivative of $f'(x)$ read “ f double-prime x .”

Note that there are no new rules to learn here. Just find the first derivative and then differentiate again.

Examples: Find the second derivative

(i) $2x^3 + \frac{1}{x} + 7$

(ii) $y = (x^2 + 1)^5$

A more applied example. Suppose that a 5-year projection of population trends suggests that t years from now, the population of a certain community will be $P(t) = -t^3 + 9t^2 + 48t + 200$.

(i) At what rate will the population be growing 3 years from now?

$$\frac{dP(t)}{dt} = -3t^2 + 18t + 48 \Big|_{t=3} = -3(3)^2 + 18(3) + 48 = 75$$

(ii) At what rate will the rate of population growth be changing with respect to time 3 years from now?

$$\frac{d^2P(t)}{dt^2} = -6t + 18 \Big|_{t=3} = 0$$

Also helpful when examining how many people are getting sick. Remember “flattening the curve.”

6.1 Higher-Order Derivatives

In general, the derivative obtained from $f(x)$ after n successive differentiations is called the **n th derivative** or **derivative of order n** and is denoted by $f^{(n)}(x)$.

For any positive integer n , the n th derivative of a function is obtained from the function by differentiating successively n times. If the original function is $y = f(x)$, the n th derivative is denoted by

$$\frac{d^n y}{dx^n} \text{ or } f^{(n)}(x)$$

Examples: Find the fourth derivative of the following

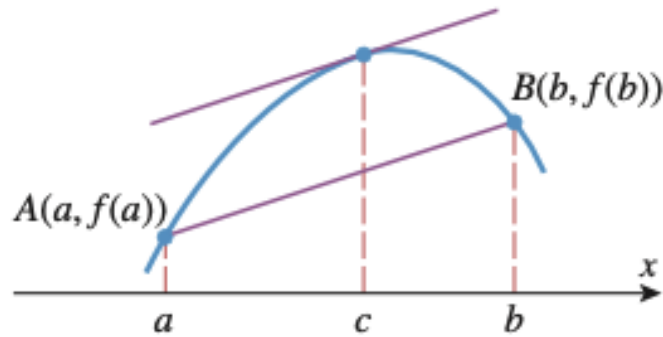
(i) $f(x) = x^4 - x^3 + 1$

(ii) $f(x) = \frac{1}{x}$

6.2 Mean Value Theorem

If f is continuous and differentiable on $[a,b]$, there exists at least one point where $x \in [a, b]$ such that the tangent line to the graph is parallel to the secant line in between these points.

Tangent Line Parallel to Secant Line



Recall that the slope of the secant line can be expressed as

$$\frac{f(b) - f(a)}{b - a}$$

. Where the slope of the tangent line, at c , is $f'(c)$.

Formally, the **Mean-Value theorem** states that if f is continuous on $[a, b]$ and differentiable on (a, b) . Then there is at least one point c in (a, b) such that

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Examples: Show that the mean-value theorem is satisfied for $f(x) = x^3 - x$ over the interval $[0, 2]$.

Find slope of secant line: Need two points $(0, f(0))$ and $(2, f(2))$. $f(0) = 0$ and $f(2) = 6$

The slope of secant line = $\frac{6-0}{2-0} = 3$.

Now we will find the point c such that

$$f'(c) = \frac{f(b) - f(a)}{b - a} = \frac{f(2) - f(0)}{2 - 0}$$

.

Find $f'(x)$

$$f'(x) = 3x^2 - 1 = 3 \implies 3x^2 = 4 \implies x = \sqrt{\frac{4}{3}} \in [0, 2] \quad \checkmark$$

7 Graphing

One way to visualize the behavior of a function of a single variable is to graph it. There are a number of methods for determining the characteristics of a graph.

Determining whether f is increasing or decreasing

- $f(x)$ is *increasing* at any point where $f'(x) > 0$
- $f(x)$ is *decreasing* at any point where $f'(x) < 0$
- $f(x)$ is *stationary* at any point where $f'(x) = 0$

Procedure for finding intervals where f is increasing or decreasing

1. Solve for $f'(x) = 0$ to determine the intervals of increase/decrease
2. Plug in test numbers into $f(x)$ on each interval to determine if the function is increasing or decreasing over that interval

Example: Find the intervals of increase and decrease

(i) $f(x) = 2x^3 + 3x^2 - 12x - 7$

$$f'(x) = 6x^2 + 6x - 12$$

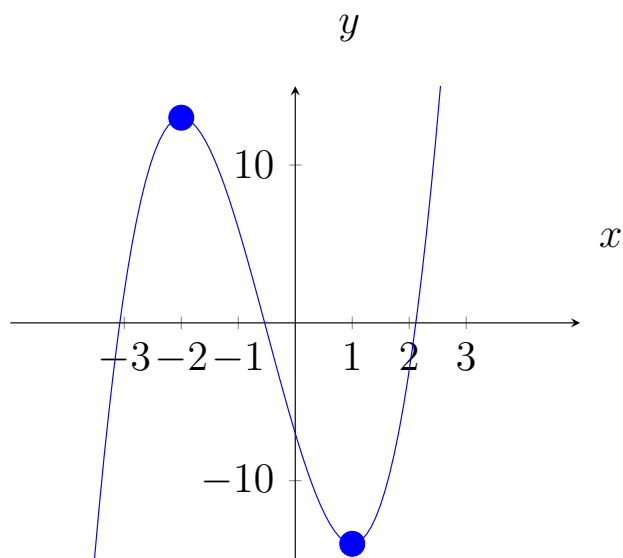
f' is defined for all numbers. Set $f'(x) = 0$. These are going to be stationary points. The function will change direction at these points.

$$6x^2 + 6x - 12 = 0 \implies x^2 + x - 2 = (x - 1)(x + 2) = 0 \implies x = -2, x = 1$$

We could try and test numbers

$$f(-2) = 13 ; f(1) = -14$$

This gives you the places where the curve will change direction. Plot these points. It is clear that they must be connected. The function at $(-2, 13)$ must fall for it to be a function analogous for $(1, -14)$.

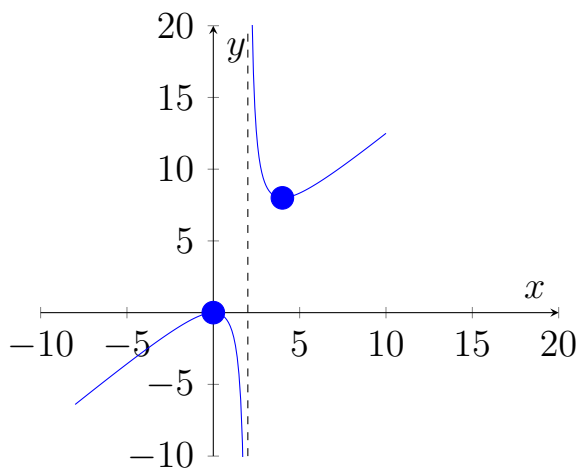


(ii) $f(x) = \frac{x^2}{x-2}$

$$f'(x) = \frac{2x(x-2) - x^2}{(x-2)^2} = \frac{x^2 - 4x}{(x-2)^2} = 0 \implies x = 0, x = 4$$

$$f(0) = 0, f(4) = 8$$

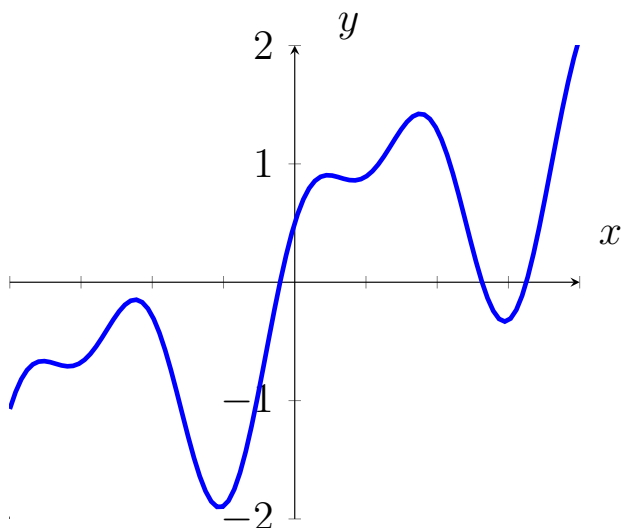
Try some points: $f(-1) = \frac{-1}{3}$, $f(1) = -1$, $f(3) = 9$, $f(5) = \frac{25}{3}$



The function is **increasing** from $(-\infty, 0)$ and $(4, +\infty)$ and **decreasing** from $(0, 2)$ and $(2, 4)$

7.1 Relative Extrema

Let's consider a function with various peaks and valleys.



Notice that there are various peaks and valleys. Calculus can help us locate these points.

A “peak” on the graph of a function is known as a **relative maximum**, and a “valley” is known as a **relative minimum**. Relative maxima and minima are collectively called **relative extrema**.

Formally, the graph of the function $f(x)$ is said to have a *relative maximum* at $x = c$ if $f(c) \geq f(x)$ for all x in an interval $a < x < b$ containing c . Similarly, the graph has a *relative minimum* at $x = c$ if $f(c) \leq f(x)$ on such an interval.

But how do we know for which values a function has a relative extremum?

If f has an extremum at x and f' exists there then $f'(x) = 0$. Of course, f' need not

exist there.³

Points that satisfy either of these conditions are called **critical numbers**.

Formally, a number c in the domain of $f(x)$ is called a **critical number** if either $f'(c) = 0$ or $f'(c)$ does not exist. The corresponding point $(c, f(c))$ on the graph of $f(x)$ is called a **critical point** for $f(x)$.

This leads us to the **first derivative test for relative extrema**:

Let c be a critical number for $f(x)$ and f be differentiable in an interval centered at c . Then the critical point $P(c, f(c))$ is:

- A **relative maximum** if $f'(x) > 0$ to the left of c and $f'(x) < 0$ to the right of c
- A **relative minimum** if $f'(x) < 0$ to the left of c and $f'(x) > 0$ to the right of c
- **Not a relative extremum** if $f'(x)$ has the same sign on both sides of c

Example: Using the first derivative test, find all critical numbers for the function $f(x) = x^4 + 8x^3 + 16x^2 - 8$ and classify each as relative max, min, or neither.

$$f'(x) = 4x^3 + 24x^2 + 32x \implies 4x(x^2 + 6x + 8) \implies 4x(x + 2)(x + 4) = 0$$

$$x = 0, x = -2, x = -4 \implies f(0) = -8, f(-2) = 8; f(-4) = -8$$

Thus, the critical points are

$$(x, y) = (0, -8), (-2, 8), (-4, -8)$$

x	-5	-4	-3	-2	-1	0	1
$f'(x)$	-60	0	12	0	-12	0	60

³For situations where there are discrete jumps, openings, or kinks, we cannot find the derivative. For kinks, recall that a derivative arises from two-sided limits. For a kink, the two-sided limits may not be equal.

The first derivative test will always give us the nature of the relative extrema. However, there is a shortcut that works much of the time. This is the **second derivative test**.

Second Derivative Test

Suppose that $f'(a) = 0$ and f'' exists and is continuous in an interval centered around a , then:

- Suppose $f''(a) > 0$, then f has a relative minimum at $x = a$
- If $f''(a) < 0$, then f has a relative maximum at $x = a$

However, if $f''(a) = 0$ or if $f''(a)$ does not exist, the test is inconclusive.

Limitation of the second derivative test

- Sometimes computing the second derivative is messy
- The test only applies to critical points at which the first derivative is zero, and not to those where the derivative is undefined.
- If both $f'(c) = 0$ and $f''(c) = 0$, the second derivative test is inconclusive.

In these cases, we can always use the first derivative test to find the nature of the critical point.

If both $f'(c) = 0$ and $f''(c) = 0$, then we can use another shortcut– the **successive derivative test**– to learn the nature of $(c, f(c))$.

The Successive Derivative Test

If both $f'(c) = 0$ and $f''(c) = 0$, then $(c, f(c))$ is:

- An **inflection point** if the first nonzero value of a higher-order derivative is an **odd-numbered derivative**. An inflection point of a continuous function is a point where the function changes from “cap-like” to “cup-like” or vice-versa.
- A **relative maximum** if the first nonzero value of a higher-order derivative is an **even-numbered derivative** with a negative value.
- A **relative minimum** if the first nonzero value of a higher-order derivative is an **even-numbered derivative** with a positive value.

Example: Find relative extrema for the following functions

(i) $f(x) = -7x^2 + 126x - 23$

(ii) $f(x) = x^3 - 12x^2 + 36x + 8$

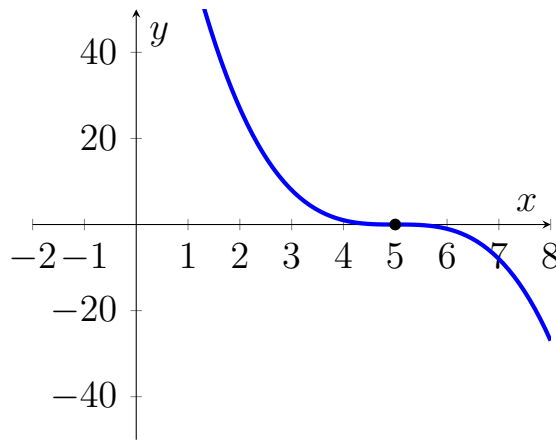
(iii) $y = (5 - x)^3$

$$\frac{dy}{dx} = 3(-1)(5 - x)^2 \implies x = 5$$

$$\text{Check } x = 5 \text{ in } y'' \implies y''(x) = -6(-1)(5 - x) \Big|_{x=5} = 0$$

This means the second-derivative test is inconclusive. We can try the first derivative test. However, $y'(4) < 0$ and $y'(6) < 0$.

If we take another derivative, we get $y''' = -6$. This is the first nonzero value of a higher-order derivative, and it is an **odd-numbered derivative** (3). Thus, this is an **inflection point**.



Note that this is neither a relative maximum nor minimum. Here, we get a saddle point or a critical point of a function where the function's derivative is zero, but it is neither a relative minimum nor a relative maximum. At a saddle point, the function changes concavity, and the nearby points on one side of the saddle point are higher, while the nearby points on the other side are lower.

(iv) $y = -(x - 8)^4$

(v) $y = -2(x - 6)^6$ $y^{(6)} = -1440$. So, the first nonzero value of a higher-order derivative is an **even-numbered** derivative with a negative value $\implies$ a relative maximum. You could have also used the first derivative test.

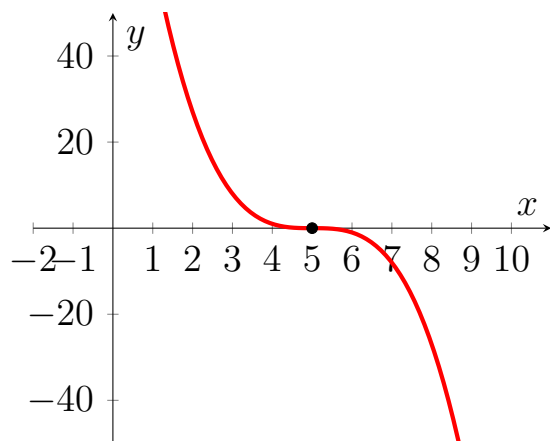
7.2 Concavity and Convexity

The following notions of **concavity** and **convexity** describe the increase and decrease of the slope of the tangent to a curve.

Definition: If the function $f(x)$ is differentiable on the interval $a < x < b$, then the graph of f is:

- **Convex** on $a < x < b$ if $f'(x)$ is increasing on the interval (“cup-like” or hold water)
- **Concave** on $a < x < b$ if $f'(x)$ is decreasing on the interval (“cap-like” or spills water)

Graphically,

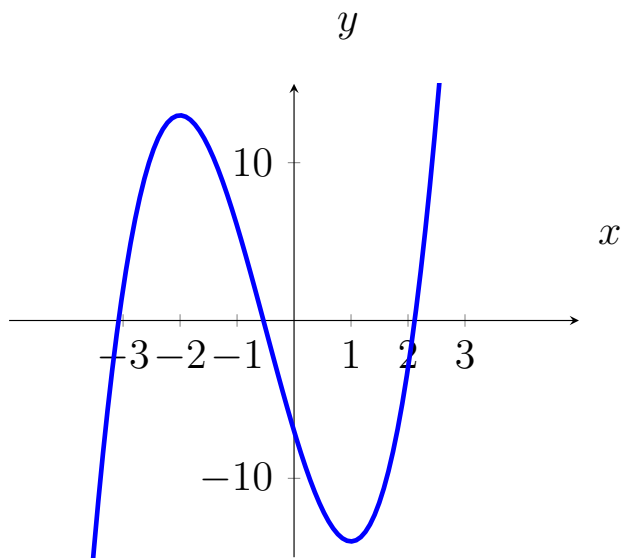


There is a simple characterization of concavity/convexity in terms of the sign of the second derivative.

Test for Concavity

- If $f''(x) > 0$ on the interval $a < x < b$, then f is convex on this interval
- If $f''(x) < 0$ on the interval $a < x < b$, then f is concave on this interval

Do not confuse concavity/convexity of a curve with its increase or decrease! A curve that is convex on an interval may be either increasing or decreasing on that interval. The same goes for concavity.

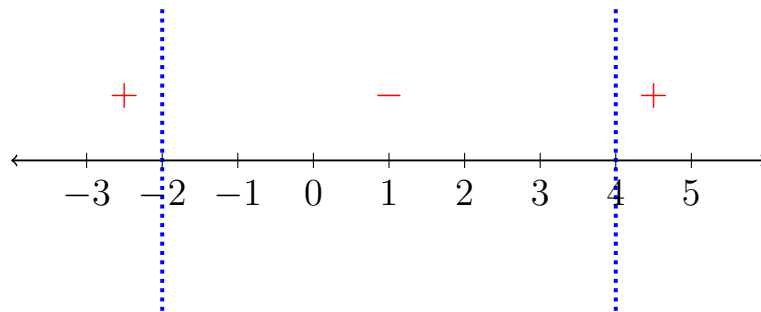


Example: Determine where the function $f(x) = x^3 - 3x^2 - 24x$ is increasing and decreasing, and where its graph is concave and convex.

Calculate the first two derivatives:

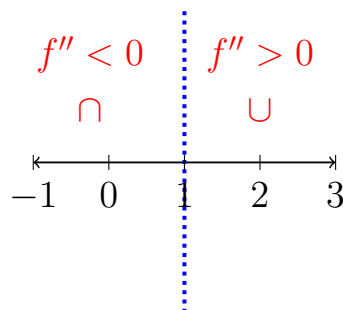
First $\rightarrow$

$$f'(x) = 3x^2 - 6x - 24 = 0 \implies 3(x^2 - 2x - 8) = 3(x-4)(x+2) = 0 \implies x = 4, x = -2$$



Second $\rightarrow$

$$f''(x) = 6x - 6 = 0 \implies x = 1$$



Recall that a point on the graph of a function $f(x)$ where the concavity changes is called an **inflection point**.

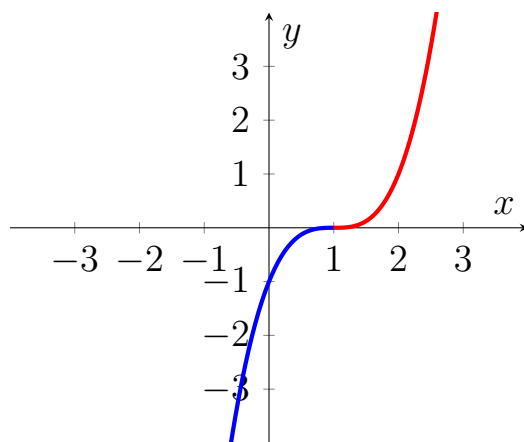
At an inflection point, either $f''(c) = 0$ or $f''(c)$ does not exist. The sign of the first derivative is immaterial.

However, just because $f''(c) = 0$, you cannot necessarily conclude that $(c, f(c))$ is an inflection point. For example, $f(x) = x^4$, then $f''(x) = 12x^2$, and the graph of f

is always concave up, even though $f''(0)$.

Determining Concavity and Locating Inflection Points

- Find f'' and determine numbers where $f''(x) = 0$ or $f''(x)$ does not exist
- Mark the numbers on a number line, dividing the line into sub-intervals. Evaluate $f''(p)$ for a test number p in each subinterval.
- The graph is convex on those intervals where $f''(p) > 0$ and concave where $f''(p) < 0$. The inflection points are the subdivision points where the concavity changes.



Here the inflection point is at $x=1$.

General Procedure for Sketching a Graph

Step 1: Find the domain of $f(x)$ (that is, where $f(x)$ is defined)

Step 2: Find and plot all intercepts. The y -intercept (where $x = 0$) is usually easy to find, but the x -intercepts (where $f(x) = 0$) may require a calculator.

Step 3: Determine all vertical and horizontal asymptotes of the graph. Draw the asymptotes in a coordinate plane

Step 4: Find $f'(x)$ and use it to determine the critical number of $f(x)$ and intervals of increase and decrease

Step 5: Determine all relative extrema (both coordinates). Plot each relative maximum with a “cap” and each relative minimum with a “cup.”

Step 6: Find $f''(x)$ and use it to determine intervals of concavity/convexity and inflection points. Plot each inflection point with a “twist” to suggest the shape of the graph near the point.

Step 7: Connect points/lines on the plot.

Example: Draw the graph $f(x) = \frac{x}{(x+1)^2}$

Step 1: Domain all $x \neq -1 \implies$ discontinuity

Step 2: Plot intercepts.

$$\frac{x}{(x+1)^2} = 0 \implies x = 0 \text{ For x-intercept}$$

$$\frac{0}{(0+1)^2} = y \implies y = 0 \text{ for y-intercept}$$

Step 3: Asymptotes

For vertical asymptotes, we see that

$$\lim_{x \rightarrow -1} \frac{x}{(x+1)^2} = -\infty$$

For horizontal asymptotes,

$$\lim_{x \rightarrow \pm\infty} \frac{x}{(x+1)^2} = 0$$

Step 4: Find critical numbers and intervals of increase and decrease

$$f'(x) = \frac{(x+1)^2 - 2x(x+1)}{(x+1)^4} = 0$$

$$\begin{aligned} \implies x^2 + 2x + 1 - 2x^2 - 2x &= 0 \implies -x^2 + 1 = 0 \implies x = \pm 1 \\ \implies x = 1, f(1) &= \frac{1}{4} \end{aligned}$$

$$f'(.5) > 0, f'(2) < 0 \implies \uparrow \in (-1, 1) \quad \downarrow \in (1, \infty)$$

Step 5: Relative max at $(1, \frac{1}{4})$

Step 6: Find $f''(x)$ to determine concavity and inflection points

$$f''(x) = \frac{-2x(x+1)^4 - 4(x+1)^3(-x^2+1)}{(x+1)^8} = 0$$

$$\implies (x+1)^3(-2x^2 - 2x + 4x^2 - 4) = 0$$

$$\implies 2(x+1)^3(x^2 - x - 2) = 0 \implies x = 2$$

Determine concavity $f''(1) < 0, f''(3) > 0$. The graph is concave on $(-\infty, -1)$ and on $(-1, 2)$ (recall $x = -1$ is not in the domain) and convex on $(2, \infty)$

Step 7: Plot the figure

